

Harry Greatorex | PhD

Research Scientist

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SKILLS

Programming & Tools: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch, Matplotlib, SunPy, Astropy, h5py, xarray), SQL (pgSQL, pgAdmin), Linux/UNIX, Bash, Git, IDL (SolarSoft), LaTeX

Data & ML: Large-scale data analysis, data modelling, feature extraction, supervised classification, statistical analysis, data visualisation

Systems: Pipeline development, automated workflows, HPC, parallel compute, remote compute environments, reproducible analysis

Data Quality & Efficiency: Instrument calibration and cross-calibration, multi-source validation, signal integrity assessment, data volume reduction and streamlining

AI Workflows: Effective use of LLMs for system development, data exploration, and automation

Leadership: Project conceptualisation and ownership, mentoring and supervision, conference chairing/organisation

Communication: Scientific writing (peer-reviewed publications), conference speaking, invited seminar talks

ADDITIONAL EXPERIENCE

Team Coordination

Managed a recreational football team over multiple seasons, coordinating logistics, sponsorships, schedules, and communications.

Workplace Inclusivity

Participated in workplace initiatives on diversity and equity, supporting an inclusive professional environment.

Public Engagement

Delivered invited talks to amateur groups, organised community events including NI Science Festival, provided software classes for local schools as part of Pathways programme.

PROFILE

Computationally focused scientific researcher with over five years of experience in Python programming, experimental data analysis, automated data workflows, data visualisation, and machine learning. Experienced in high-performance computing, pipeline development, and statistical analysis of large, complex datasets. Strong record of leading projects, mentoring junior researchers, and delivering practical insights through automation and advanced analytics. Engaged in cross-disciplinary collaborations and outreach programmes, with a clear commitment to inclusive working practices. Motivated to apply strong technical, analytical, and problem-solving skills within collaborative teams, bringing a pay-it-forward mindset that prioritises shared success.

EXPERIENCE

Postdoctoral Research Fellow

2025–Present

Astrophysics Research Centre, Queen's University Belfast

Data Pipelines & Automation

- Designed, built, and maintained **end-to-end Python pipelines** for large-scale data analysis, covering ingestion, cleaning, feature extraction, validation, and structured outputs across **multi-terabyte datasets**.
- Reduced manual analysis overhead by developing reusable, automated tooling to support rapid reprocessing, parameter sweeps, and reproducible workflows.

Machine Learning & Analytics

- Developed and deployed **supervised machine-learning models** (PyTorch, MLPs) to classify weak signals in noisy, real-world datasets, combining automated detection with informed review to improve reliability.
- Performed statistical analysis across significant data volumes, applying robust data cleaning, feature engineering, and evaluation techniques to classify and parametrise observational data.

Systems, Scale & Infrastructure

- Scaled analytical workflows across **high-performance computing environments** using parallel execution and GPU-enabled remote systems to efficiently process large datasets.
- Worked extensively in remote Linux/Unix environments, managing data transfer, job execution, and shared compute resources.

Project Leadership & Collaboration

- Led an independent, data-driven research programme** from concept to delivery planning, including preparation of a competitive £250k funding proposal that achieved a strong initial review score (>80%).
- Recruited and supervised a research intern, defining technical scope and milestones and mentoring them through a project that produced publishable results.
- Collaborated across international, cross-disciplinary teams and contributed to mission-focused working groups and review meetings.

Communication & Engagement

- Presented complex technical results to specialist and non-specialist audiences through talks, reviews, and collaborative meetings.
- Chaired sessions at an international conference, supporting discussion, coordination, and community engagement.

Sales Assistant

2015–2020

Millets, Derby

- Worked in a **KPI-driven retail environment**, delivering customer-focused sales, specialist product advice, and consistent performance against individual and team targets.

EDUCATION

Doctor of Philosophy (PhD) in Astrophysics 2021–2025

Queen's University Belfast

Thesis: “Observational Analysis of Lyman-Alpha Emission in Solar Flares”

- Investigated high-energy solar phenomena using complex spectroscopic, photometric, and image-based analyses of space-based mission data, utilising bespoke computational tools within the SolarSoftware framework for IDL and SunPy for Python.
- Performed detailed instrument performance evaluations across multiple space missions, requiring in-depth knowledge of sensor architecture, calibration workflows, and precision measurement techniques.
- Secured competitive funding to undertake an overseas research placement at **NASA Goddard Space Flight Center** as a visiting research scholar.
- Served as a **Chief Observer** within the Max Millennium solar forecasting programme, integrating and interpreting multi-source observational data.

MSci in Physics (First Class Honours)

2017–2021

Queen's University Belfast

Final-Year Project: “Computational Modelling of the Chromospheric Response to Solar Flare Heating”

- Specialised in astrophysics with additional studies in plasma physics, medical physics, and materials science.
- Completed a final-year research project in radiative hydrodynamic modelling of the solar atmosphere using RADYN and the FCHROMA grid, run entirely in a remote computing environment via Citrix systems.
- Collaborated with supervisors through regular virtual progress meetings, developing strong communication and self-management skills in a remote research setting.